Hey Ting, the goal is to keep the system as **low-complexity, low-cost**, and thin (<1.5 mm) as possible, suitable for initial prototyping and validation. Below is the current architecture being considered, along with component options and compatibility:

MCU + BLE Options

Option	Description	Notes
A	**nRF52832-QFAA-R** (WLCSP) + Trace Antenna	Requires external 32 MHz & 32.768 kHz crystals. Needs RF layout and BLE cert.
B	**nRF52832-**	Same as A but

	QFAA-R + **Johanson 2450AT18A100** Chip Antenna	uses chip antenna. Still needs BLE cert.
C	**DA14531MOD-00F01002** + External Flash (e.g., W25X05CLUXIG)	Built-in antenna and 32 MHz crystal. Needs external SPI Flash. BLE certified.
D	**MDBT42V-512KV2** (nRF52832 Module)	Fully integrated, BLE certified. But

	~2.5 mm thick (too thick for form factor).

Core Sensors + Components (compatible with all options unless noted)

- **PPG Sensor**: MAX30102.
- **IMU**: LSM6DSOXTR - single unit, no longer separate components as per original spec.
- **Barometer**: LPS22HHTR - new addition.
- **Temperature Sensor:** No longer applicable.
- **PMIC**: BQ25120AYFPT.
- **Flash**: Only needed for **Option C** (e.g., W25X05CLUXIG).
- **Battery:** LPCV170726.
- **Crystals**:
  - 32 MHz (e.g., TXC 9HT30-32.000M) – required for Options A & B.
  - 32.768 kHz (e.g., Abracon ABS07) – optional RTC crystal.

Design Considerations

- Form factor: Max height <1.5 mm (MDBT42V form factor is not ideal).
- Preference for **certified**

modules (Options C/D) to simplify RF compliance.
- This is strictly a **Proof-of-Concept**, so minimal BOM and simplified design is the priority.

Let me know if you can help with:

- Architecture recommendation for using Nordic vs Dialogue MCU given the overall goal.
- Recommending layout or integration best practices.
- Schematic, layout and firmware for this PoC .